



ENTERPRISE ENTITY-RELATIONSHIP (ER) ARCHITECTURE & RELATIONAL SCHEMA SPECIFICATION

Comprehensive Logical Data Model, Multi-Level Cascading & Frappe Backend Blueprint

Document Version:	v3.0 Production Baseline (Approved Post-Review Architecture)
Target Framework:	Frappe Framework (Python 3 / MariaDB Relational Database)
Requirements Baseline:	SRS v3.0, Adama Workshop (Sep 2026), KPI Directives & Production Code
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Executive Summary & Architectural Scope

This document establishes the authoritative, enterprise-grade Entity-Relationship (ER) model and physical schema specification for the Ethiopian Red Cross Society (ERCS) PMER-MIS platform. The system replaces fragmented Excel-based tracking with a unified, web-based, multi-tenant-by-branch Management Information System powered by Frappe.

The data architecture synthesizes the 5-Year Strategic Plan (2025-2030), the Results-Chain causal logic model (Activity -> Output -> Outcome -> Impact) with standardized coding (e.g., EPR-OUT-01), dual-stream Annual Operational Plans (AOP-CORE Hamle-Sene vs AOP-Donor MOU timelines), the Region Activity Link associative gateway for Branch Head governance, project contributing vs standalone non-contributing activities, financial Maker-Checker sign-offs, and multi-vault Knowledge Management.

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1. Conceptual Model & Disciplined Relational Conventions

1.1 Disciplined Component Conventions

- Entities (Rectangles):**
Named with singular, capitalized nouns (e.g., Project, Plan Entry). Strong Entities possess independent lifecycles and natural/synthetic primary keys (Standard DocTypes). Weak Entities depend existentially on a parent entity (Submittable Child Tables).
- Attributes (Compartments):**
Listed inside entity boxes in Crow's Foot table style. Simple atomic values, Multi-valued attributes resolved into child tables (1NF), and Derived metrics (achievement %, burn rate, dependency ratio) computed dynamically.
- Keys (PK / FK):**
Every entity without exception has an explicit Primary Key (PK). Foreign Keys (FK) establish rigorous relational links. In Frappe, PKs map to name/autoname, and FKs map to Link fields with database indices.
- Relationships & Verbs:**
Active verb phrases describe exact operations (e.g., 'allocates', 'monitors', 'cascades into'). Every relationship connects exactly two entities with no dangling links.
- Cardinality & Participation:**
Expressed using Information Engineering Crow's Foot notation: One-to-One (1:1), One-to-Many (1:N), Many-to-Many (M:N). Total Participation (mandatory, double bar/solid line) vs Partial Participation (optional, circle/dashed line).

1.2 Structural Integrity & Normalization Proofs

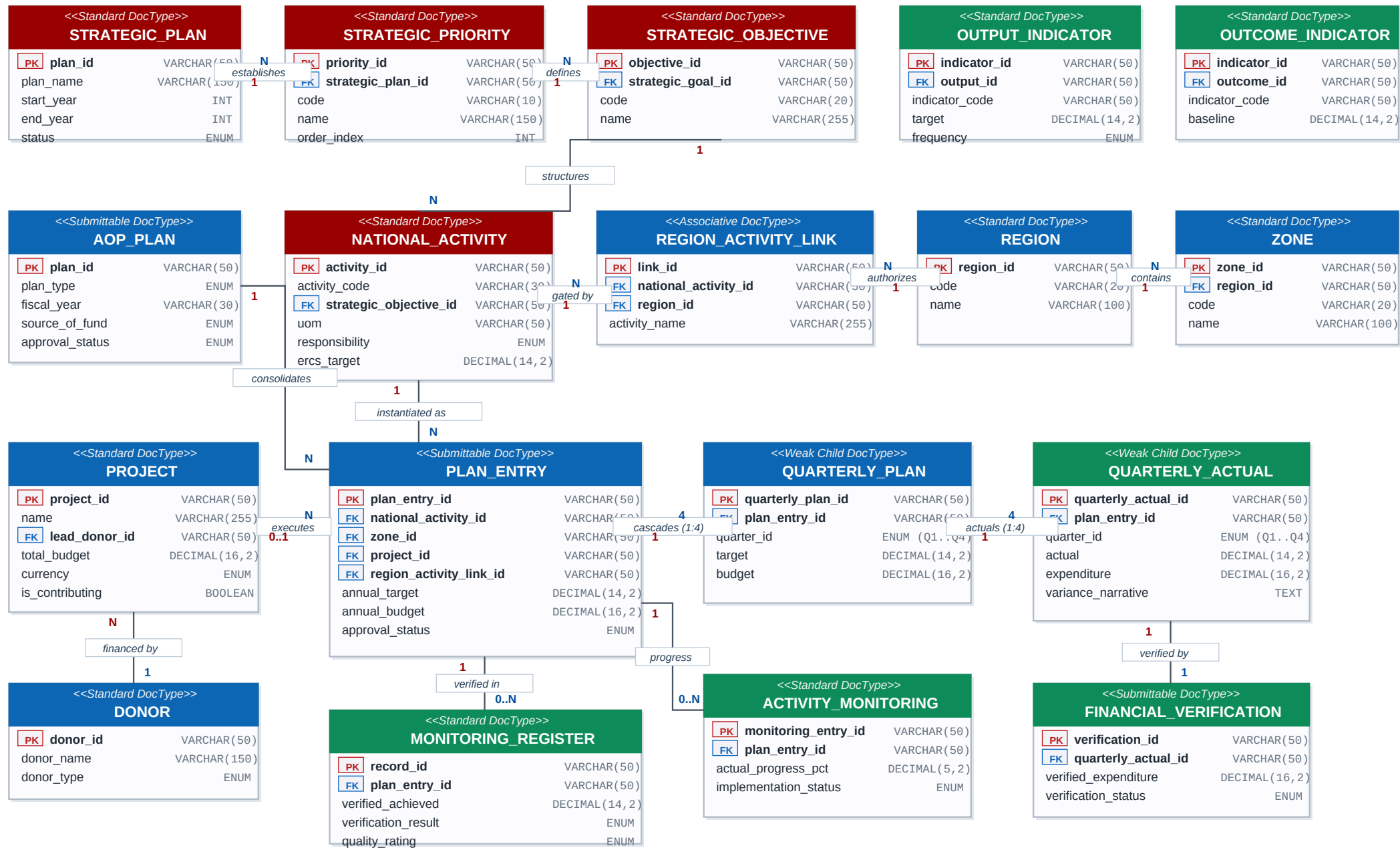
- First Normal Form (1NF):**
All attribute values are strictly atomic. Repeating groups (such as quarterly targets, multi-donor contributions, and demographic splits) are normalized into child entities (Quarterly Plan, Project Donor Allocation, Disaggregation Entry).
- Second Normal Form (2NF):**
All non-key attributes are fully functionally dependent on the primary key. Composite keys in junction entities (e.g., Region Activity Link) carry attributes strictly relevant to the associative relationship.
- Third Normal Form (3NF):**
All transitive dependencies are eliminated. Master references (Strategic Priorities, UOM Multipliers, Status Thresholds) exist independently, preventing update anomalies.
- M:N Resolution:**
All Many-to-Many relationships are formally resolved into Associative Junction Entities (e.g., Region Activity Link, Project Donor Allocation) equipped with their own audit metadata.

1.3 How to Read the Vector ER Diagrams

Entity Header Color Codes:		Connector & Cardinality Symbols:	
	Strategic Framework (Pillars, Objectives, Master Activities)		Exactly 1 to Exactly 1 (Mandatory One-to-One)
	Operational Planning & Cascading (AOP, Projects, Regions, Plan Entries)		Exactly 1 to 1 or More (Mandatory One-to-Many, Total Participation)
	Execution, Indicator Monitoring & Verification (Quarterly Actuals, Registries)		Exactly 1 to 0 or More (Optional One-to-Many, Partial Participation)
	Governance, Knowledge Management & Workflows (Clubs, Vaults, Audits)		0 or 1 to 0 or More (Conditional Associative Linkage)
		[PK] / [FK]	Primary Key (Red Badge) / Foreign Key (Blue Badge)

Diagram 1: PMER-MIS Enterprise Master Architecture ERD (End-to-End Core Backbone)

Illustrates the complete operational backbone from 5-Year Strategy down to Zonal/Project Execution, Quarterly Actuals, and Verification.



Models the 5-Year Results-Based Management (RBM) causal hierarchy: Activity -> Output -> Outcome -> Impact alongside Standalone Strategic KPIs.

Illustrates dual-stream AOP plans, Regional gateway (Region Activity Link), Project-to-AOP alignment, and non-programmatic budget lines.



Diagram 4: Quarterly Execution Cycles, Monitoring Registers & Field Verification ERD

Details quarterly transactions, high-frequency monthly indicator values with demographic disaggregation, dual monitoring, and finance verification.

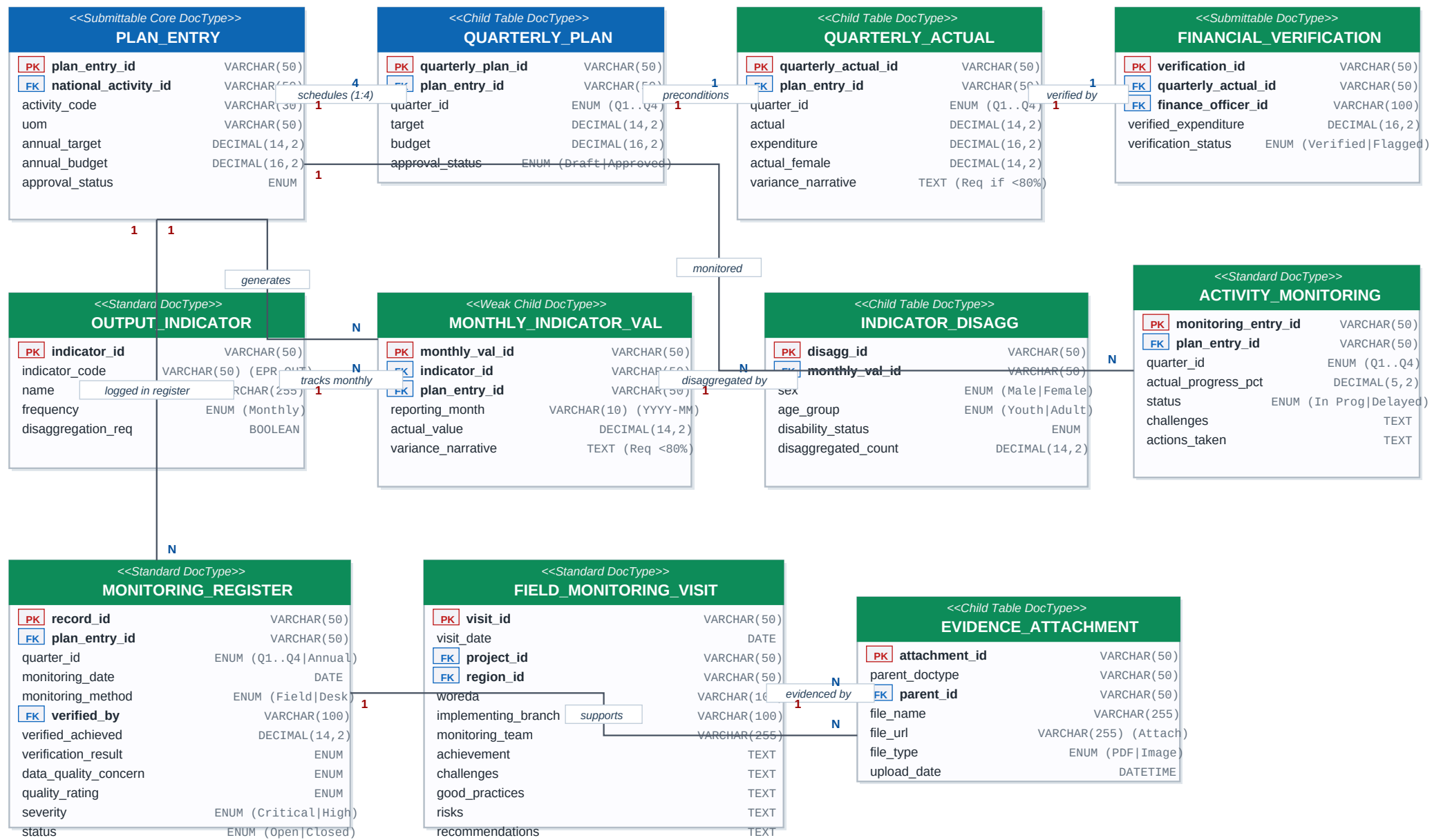


Diagram 5: Governance, Knowledge Management Vaults & Workflow Engine ERD

Models explicit Volunteer vs Member separation, Red Cross Clubs, the 6 KM Institutional Memory Vaults, and the Multi-Level Approval Engine.



2. Comprehensive Entity Catalog & Business Roles

Tabular catalogue of all 32 entities defining Frappe DocType types, classification, and business roles.

Entity Name	Module	Type	Frappe DocType	Business Role / Functionality	PK Field
Strategic Plan	Module 1	Strong	Standard DocType	Macro 5-year institutional plan parameters...	plan_id
Strategic Priority	Module 1	Strong	Standard DocType	Core strategic pillars (Disaster Prep, Hea...	priority_id
Strategic Goal	Module 1	Strong	Standard DocType	Institutional goal statements subordinate ...	goal_id
Strategic Objective	Module 1	Strong	Standard DocType	Measurable 5-year targets supporting Strat...	objective_id
National Activity	Module 1	Strong	Standard DocType	Master 284+ pre-approved activity menu wit...	activity_id
Output & Indicator	Module 1	Strong	Standard DocType	Immediate deliverables and standardized in...	indicator_id
Outcome & Indicator	Module 1	Strong	Standard DocType	Intermediate institutional/community chang...	indicator_id
Impact & Indicator	Module 1	Strong	Standard DocType	Long-term societal impact indicators (e.g....	indicator_id
Strategic KPI	Module 1	Strong	Standard DocType	Standalone 5-year milestone KPIs with stat...	kpi_id
KPI Progress Entry	Module 1	Weak	Child Table	Mid-Year and End-Year evaluated KPI progre...	progress_id
Region	Module 2	Strong	Standard DocType	14 Regional Branches overseeing geographic...	region_id
Zone	Module 2	Strong	Standard DocType	40+ Zonal Branches executing field operati...	zone_id
Woreda	Module 2	Strong	Standard DocType	District-level operational and community i...	woreda_id
Department	Module 2	Strong	Standard DocType	Technical and operational departments (con...	department_id
AOP Plan Header	Module 2	Strong	Submittable	Dual-stream annual plans (AOP-CORE vs AOP-...	plan_id
Project	Module 2	Strong	Standard DocType	Donor-funded interventions (ETB, EUR, USD ...	project_id
Donor	Module 2	Strong	Standard DocType	Grant partners (IFRC, ICRC, PNS, USAID, lo...	donor_id
Project Donor Alloc	Module 2	Weak	Child Table	Multi-donor split & local vs donor depende...	allocation_id
Region Activity Link	Module 2	Strong	Associative	Branch Head governance gateway: allocates ...	link_id
Non-Programmatic Act	Module 2	Strong	Standard DocType	Overhead operations (Legal, Supply Chain, ...	npa_id
Plan Entry	Module 2	Strong	Submittable	Universal operational planning atomic unit...	plan_entry_id
Quarterly Plan	Module 3	Weak	Child Table	Q1-Q4 cascaded planned targets and budgets...	quarterly_plan_id
Quarterly Actual	Module 3	Weak	Child Table	Q1-Q4 reported achievements and expenses (...)	quarterly_actual_id
Monthly Ind Value	Module 3	Weak	Child Table	High-frequency monthly indicator data entr...	monthly_val_id
Indicator Disagg	Module 3	Weak	Child Table	Sex, age group, disability, and vulnerabil...	disagg_id
Monitoring Register	Module 3	Strong	Standard DocType	Structured PMER verification (ratings, sev...	record_id
Field Monitoring Visit	Module 3	Strong	Standard DocType	Qualitative inspection visits capturing na...	visit_id
Evidence Attachment	Module 3	Weak	Child Table	PDF and photographic evidence documents va...	attachment_id
Financial Verification	Module 3	Strong	Submittable	Finance Officer Maker-Checker expenditure ...	verification_id
Red Cross Club	Module 4	Strong	Standard DocType	Youth school, university, and community cl...	club_id
Volunteer Record	Module 4	Strong	Standard DocType	Operational human resources (skills, activ...	volunteer_id
Member Record	Module 4	Strong	Standard DocType	Governance stakeholders with voting rights...	member_id

3. Field-Level Data Dictionary -- Core Architecture

Authoritative field-level schema definitions, MariaDB storage types, Frappe types, and business validation rules.

Table 1: National Activity (`tabNational Activity`) -- Master Catalog

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
activity_id	VARCHAR(50)	Data (Name)	PK Not Null	Unique synthetic identifier (e.g. 'NA-1.1.1').
activity_code	VARCHAR(30)	Data	Unique Not Null	Official hierarchical code (e.g. 1.1.1).
strategic_objective_id	VARCHAR(50)	Link: Strategic Objective	FK Not Null	Foreign key to parent Strategic Objective.
name	VARCHAR(255)	Data	Not Null	Master activity title from 2025-2030 framework.
uom	VARCHAR(50)	Link: UOM Conversion	FK Not Null	Unit of measurement (Household, Kit, Person).
responsibility	VARCHAR(50)	Select	Not Null	HQ, Branch, Both, or named division.
ercs_target	DECIMAL(14, 2)	Float	Nullable	Seeded national target figure (Authoritative bas...
ercs_budget	DECIMAL(16, 2)	Currency (ETB)	Nullable	Seeded national budget figure in Ethiopian Birr.

Table 2: Plan Entry (`tabPlan Entry`) -- Universal Operational Unit

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
plan_entry_id	VARCHAR(50)	Data (Name)	PK Not Null	Atomic operational planning commitment ID.
plan_id	VARCHAR(50)	Link: AOP Plan	FK Nullable	Parent AOP plan header (AOP-CORE or AOP-Donor).
scope_type	VARCHAR(30)	Select	Not Null	Regional, Project, NonProgrammatic.
national_activity_id	VARCHAR(50)	Link: National Activity	FK Nullable	Linked master activity. Mandatory if is_contribu...
zone_id	VARCHAR(50)	Link: Zone	FK Nullable	Executing Zone (mandatory for Regional scope).
project_id	VARCHAR(50)	Link: Project	FK Nullable	Executing Project (mandatory for Project scope).
region_activity_link_id	VARCHAR(50)	Link: Region Activity Link	FK Nullable	Branch Head gateway link authorizing execution.
is_contributing	SMALLINT	Check	Not Null	1 = sums into National Society Total; 0 = Quaran...
annual_target	DECIMAL(14, 2)	Float	Not Null	Planned annual target. If 0, budget locked to 0.
annual_budget	DECIMAL(16, 2)	Currency	Not Null	Annual budget. Inputs locked until target entere...
target_female	DECIMAL(14, 2)	Float	Nullable	Target female beneficiary breakdown.
community_type	VARCHAR(50)	Select	Nullable	Host Community, IDP, Refugee, Returnee, Mixed.
approval_status	VARCHAR(30)	Select	Not Null	Draft, Pending Approval, Approved, Rejected.

Table 3: Quarterly Actual (`tabQuarterly Actual`) -- Execution Transactions

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
quarterly_actual_id	VARCHAR(50)	Data (Name)	PK Not Null	Quarterly achievement record identifier.
plan_entry_id	VARCHAR(50)	Link: Plan Entry	FK Not Null	Parent Plan Entry (Cascade delete on removal).
quarter_id	VARCHAR(10)	Select	Not Null	Q1, Q2, Q3, Q4.
actual	DECIMAL(14, 2)	Float	Not Null	Reported physical achievement for this quarter.
expenditure	DECIMAL(16, 2)	Currency	Not Null	Reported financial expenditure for this quarter.
actual_female	DECIMAL(14, 2)	Float	Nullable	Reported female beneficiaries reached.
variance_narrative	TEXT	Text	Nullable	Mandatory explanation if achievement < 80%.
approval_status	VARCHAR(30)	Select	Not Null	Draft, Pending Approval, Approved, Rejected.

4. Field-Level Data Dictionary -- Monitoring, Governance & KM

Table 4: Monitoring Register Record (`tabMonitoring Register Record`)

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
record_id	VARCHAR(50)	Data (Name)	PK Not Null	Monitoring register entry ID.
plan_entry_id	VARCHAR(50)	Link: Plan Entry	FK Not Null	Linked Plan Entry.
quarter_id	VARCHAR(15)	Select	Not Null	Q1, Q2, Q3, Q4, Annual.
monitoring_date	DATE	Date	Nullable	Inspection or verification date.
monitoring_method	VARCHAR(30)	Select	Nullable	Field visit, Desk review, Remote, Joint.
verified_by	VARCHAR(100)	Link: User	FK Nullable	PMER verifier user account.
verified_achieved	DECIMAL(14, 2)	Float	Nullable	Independently verified physical metric.
verification_result	VARCHAR(30)	Select	Nullable	Fully verified, Partially verified, Not verified...
data_quality_concern	VARCHAR(30)	Select	Nullable	None, Validity, Integrity, Precision, Timeliness...
quality_rating	VARCHAR(30)	Select	Nullable	Good, Satisfactory, Needs improvement, Poor.
severity	VARCHAR(20)	Select	Nullable	Critical, High, Medium, Low.
recommendation	TEXT	Text	Nullable	Documented corrective action requirement.
status	VARCHAR(20)	Select	Not Null	Open, In Progress, Closed.

Table 5: KM Vault Report (`tabKM Vault Report`) -- Institutional Repository

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
report_id	VARCHAR(50)	Data (Name)	PK Not Null	Institutional report ID.
title	VARCHAR(255)	Data	Not Null	Mandatory official document title.
report_category	VARCHAR(50)	Select	Not Null	Assessments, Monitoring reports, Evaluation repo...
file_format	VARCHAR(20)	Select	Not Null	PDF, Word, Excel, PowerPoint.
file_url	VARCHAR(255)	Attach	Not Null	Stored file attachment URL in Frappe vault.
description	TEXT	Text	Not Null	Mandatory 50-150 word summary of findings.
program_project_name	VARCHAR(255)	Data	Not Null	Associated project or intervention title.
region_location	VARCHAR(50)	Link: Region	FK Not Null	Geographic target region.
access_level	VARCHAR(20)	Select	Not Null	Public, Internal, Restricted (Admin/PMER only).
uploaded_by	VARCHAR(100)	Link: User	FK Not Null	System-captured user account (auto-populated).

Table 6: Volunteer Record (`tabVolunteer Record`) -- Operational Capital

Field Name	SQL Type	Frappe FieldType	Key / Null	Validation & Business Integrity Rules
volunteer_id	VARCHAR(50)	Data (Name)	PK Not Null	Unique volunteer identifier.
full_name	VARCHAR(150)	Data	Not Null	Volunteer legal full name.
zone_id	VARCHAR(50)	Link: Zone	FK Nullable	Parent Zonal branch.
club_id	VARCHAR(50)	Link: Red Cross Club	FK Nullable	Associated Red Cross school/community club.
gender	VARCHAR(10)	Select	Not Null	Male, Female.
age_group	VARCHAR(20)	Select	Not Null	Youth (18-29), Adult (30+).
skills_profile	VARCHAR(255)	Data	Nullable	Specialized skills (First Aid, WASH, Logistics).
is_active	SMALLINT	Check	Not Null	1 = Active on operational roster; 0 = Inactive.

5. Relationship Cardinality & Participation Matrix

Exhaustive matrix specifying Parent, Relationship Verb, Child, Cardinality, Participation, and Delete Rules.

Parent Entity	Relationship Verb	Child Entity	Card.	Participation	Foreign Key Link	Delete Rule
Strategic Plan	establishes	Strategic Priority	1:N	Total (1..*)	Priority.strategic_plan_id	RESTRICT
Strategic Priority	defines	Strategic Goal	1:N	Total (1..*)	Goal.strategic_priority_id	CASCADE
Strategic Goal	operationalizes	Strategic Objective	1:N	Total (1..*)	Objective.strategic_goal_id	CASCADE
Strategic Objective	structures	National Activity	1:N	Total (1..*)	Activity.strategic_objective_id	RESTRICT
Strategic Objective	drives	Outcome	1:N	Total (1..*)	Outcome.strategic_objective_id	RESTRICT
Outcome	measured by	Outcome Indicator	1:N	Total (1..*)	OutcomeInd.outcome_id	CASCADE
National Activity	generates	Output	1:N	Total (1..*)	Output.activity_id	CASCADE
Output	measured by	Output Indicator	1:N	Total (1..*)	OutputInd.output_id	CASCADE
Region	contains	Zone	1:N	Total (1..*)	Zone.region_id	RESTRICT
Zone	contains	Woreda	1:N	Total (1..*)	Woreda.zone_id	RESTRICT
National Activity	gated by	Region Activity Link	1:N	Partial (0..*)	Link.national_activity_id	RESTRICT
Region	authorizes	Region Activity Link	1:N	Partial (0..*)	Link.region_id	CASCADE
Region Activity Link	allocates zone	Eligible Zone Item	1:N	Total (1..*)	Item.link_id	CASCADE
Project	funded by	Project Donor Alloc	1:N	Total (1..*)	Alloc.project_id	CASCADE
Donor	finances	Project Donor Alloc	1:N	Partial (0..*)	Alloc.donor_id	RESTRICT
National Activity	instantiated as	Plan Entry	1:N	Partial (0..*)	Entry.national_activity_id	RESTRICT
Region	hosts	Plan Entry	1:N	Partial (0..*)	Entry.region_id	RESTRICT
Zone	executes	Plan Entry	1:N	Partial (0..*)	Entry.zone_id	RESTRICT
Project	executes	Plan Entry	1:N	Partial (0..*)	Entry.project_id	RESTRICT
Plan Entry	cascades into	Quarterly Plan	1:4	Total (4..4)	QPlan.plan_entry_id	CASCADE
Plan Entry	accumulates	Quarterly Actual	1:4	Partial (0..4)	QActual.plan_entry_id	CASCADE
Plan Entry	monitored in	Monitoring Record	1:N	Partial (0..*)	Record.plan_entry_id	CASCADE
Output Indicator	tracks monthly	Monthly Ind Value	1:N	Partial (0..*)	Val.indicator_id	RESTRICT
Monthly Ind Value	disaggregated by	Indicator Disagg	1:N	Partial (0..*)	Disagg.monthly_val_id	CASCADE
Plan Entry	evaluates	Activity Monitoring	1:N	Partial (0..*)	Mon.plan_entry_id	CASCADE
Field Visit	submits evidence	Evidence Attachment	1:N	Partial (0..*)	Attach.parent_id	CASCADE
Zone	charters	Red Cross Club	1:N	Partial (0..*)	Club.zone_id	RESTRICT
Red Cross Club	enlists	Volunteer Record	1:N	Partial (0..*)	Volunteer.club_id	SET NULL
Zone	registers	Member Record	1:N	Partial (0..*)	Member.zone_id	RESTRICT
Workflow Def	configures	Workflow Step	1:N	Total (1..*)	Step.workflow_name	CASCADE
Workflow Step	records audit	Workflow Transition	1:N	Partial (0..*)	Trans.step_id	RESTRICT

6. Frappe Backend Implementation Blueprint & Workflow Engine

Technical architecture mapping the logical ER model into Frappe Framework primitives, DocTypes, and RBAC security.

6.1 DocType Categorization & MariaDB Storage

Standard DocTypes ('is_submittable=0'):

Independent master models. Examples: Strategic Priority, National Activity, Project, Region, Field Monitoring Visit. Mapped to physical MariaDB tables `tab<DocType Name>`.

Submittable DocTypes ('is_submittable=1'):

Transactional documents with strict state machines. Examples: AOP Plan, Plan Entry, Quarterly Plan, Quarterly Actual. State tracked via `docstatus` (0=Draft, 1=Submitted, 2=Cancelled).

Child DocTypes ('istable=1'):

Weak dependent lines. Linked to parent forms via standard fields `parent`, `parentfield`, `parenttype`. Examples: Eligible Zone Item, Project Donor Allocation, Indicator Disbursement.

Single DocTypes ('issingle=1'):

Global system-wide configurations stored in `tabSingles`. Examples: PMER Global Settings, Status Threshold Settings.

6.2 Role-Based Access Control (RBAC) Scoping Rules

Super Admin (ICT):

Full unrestricted national system configuration, user provisioning, server security, and master data control.

PMER HQ Admin:

National-level visibility across all branches, projects, and achievements. Indicator library owner. Final review gatekeeper.

Branch Head:

Scoped strictly to their assigned Regional Branch via Frappe User Permissions (`region == user.assigned\_region`). Authorizes Region Activity Links and approves zonal reports.

Data Entry Officer:

Zone or Project Coordinator. Limited strictly to data entry and quarterly actual reporting for their assigned scope.

Finance Officer:

Authorized to review and verify expenditure actuals, upload financial reports, and execute Maker-Checker sign-offs.

Read-only Management:

Executive dashboard view rights (HQ Programme Head, DSG, SG, Board). SG performs final executive approval.

6.3 Unified Approval Workflow & Duplicate Prevention Engine (FR-21)

- Multi-Stage Workflow Chain: Per SRS v3.0 Comment 4, the operational review sequence terminates with Secretary General (SG) approval (Zonal -> Branch -> Regional Branch -> HQ Programme Head consolidation -> PMER review -> Secretary General SG). The Governing Board receives the SG-approved national report for strategic governance oversight but does not execute approval actions.
- National Consolidation Rule (FR-21): For each Activity Code, the system automatically sums Regional Branch contributions together with HQ Donor-Funded Project contributions into a single unified National Society Actual (Region A + Region B + Project X + Project Y = National Total). Individual contributing sources remain fully traceable and visible on drill-down.
- Duplicate Prevention Engine: Detects and flags potential overlaps where two submissions report the same activity, location, and target period, preventing double counting in consolidated outputs.